

Abel Payyanickal

905-302-7128 | sabuabel2003@gmail.com | linkedin.com/in/abel-sabu-30651121a | github.com/Abel356

SUMMARY

Computer Science student specializing in **AI/ML** and **Cybersecurity** with expertise in Python, reinforcement learning, and cloud computing using **Azure** services, **Docker**, and **CI/CD** pipelines. Experienced in full-stack development with **React**, **Node.js**, and microservices architecture, combined with hands-on IT infrastructure management including Active Directory and **SCCM** deployment. Proven ability to deliver scalable **MLOps** solutions and agile development practices in enterprise environments.

EDUCATION

York University

*Bachelor in Computer Science with Specialization in **Computer Security and AI***

Toronto, ON

Sep. 2021 – Dec. 2026

EXPERIENCE

Information Technology Assistant

May 2024 – Sep. 2024

York University

Toronto, ON

- Collaborated with our coordinator, manager, and cross-functional team to streamline our ticketing system and improve workflow speed
- Implemented process improvements, enhancing system efficiency and accelerating issue resolution
- Managed device **imaging** and deployment for new PCs and laptops using **SCCM**, streamlining processes to reduce setup time while ensuring security compliance
- Overhauled asset tracking system with barcode scanning implementation, significantly improving inventory accuracy across campus departments
- Provided technical support to faculty and staff, troubleshooting Windows, Active Directory, and network connectivity issues through help desk operations
- Contributed to SCCM cloud migration initiatives by developing configuration management policies that enhanced software deployment efficiency

Client Service Representative - Team Lead

Sept. 2023 – May 2024

York University

Sept 2024 – Present

- Enhanced campus experience by welcoming visitors and managing memberships at Sport and Recreation facilities
- Delivered exceptional client service by handling inquiries, processing renewals, and coordinating with high-profile renting partners like the *Toronto Raptors*, *Team Canada (Volleyball and Basketball)*, *National Bank of Canada*
- Streamlined facility rentals by managing invoices and **collaborated** with university departments to facilitate staff **training** enrollments
- Trained and mentored new staff on membership processes, facility rentals, and client-service best practices
- **Partnered** with the Sport and Recreation coordinator and manager to **refine** and implement our customer-service RATER model
- Led regular team huddles and workshops to monitor performance metrics and drive continuous improvements

PROJECTS

Precogni – Driver Drowsiness Detector (Hack the 6ix 2025) | Python, Gemini API, Node.js, OpenCV, Whisper AI

- Tackled the safety gap of reactive fatigue alerts by prototyping a proactive driver “**sleep preventer**” during a 3-day hackathon.
- Integrated the **Gemini AI API** into a computer-vision pipeline to read early fatigue cues (eye closure, yawning, head tilt) and trigger preventive engagement.
- Built the alerting loop, tuned thresholds, and coordinated **CV + AI integration** across a 4-person team under tight time constraints.

SaveBetter – Adaptive Financial Planning App (Kiro [AWS] Hackathon) | Next.js, React 18, TypeScript, Tailwind

- Targeted shortcomings of rigid 50/30/20 rules for Canadians by delivering personalized, adaptive savings guidance at scale.

- Developed adaptive saving engine with sophisticated algorithms and **AI-enhanced** behavioral **analysis** using **OpenAI GPT-4** for personalized financial recommendations and bias detection and a dual assessment system with **machine learning** pattern recognition, confidence scoring, and real-time Canadian banking data integration from major institutions **RBC, TD, BMO**
- Engineered a production-ready framework for GPT-4 integration—env-keyed switch, typed interfaces, and prompts—enabling seamless upgrade from simulated insights to real AI analysis.

RetailAI: Microsoft Azure Hackathon | *Azure ML, Custom Vision, Azure OpenAI, Node.js, Kubernetes*

- Built retail AI platform using **Azure ML, Custom Vision** for image-based product search, and RAG-powered **chatbot** with Azure **OpenAI** for natural language inventory queries
- Developed data **pipelines** for anomaly detection, integrated structured/unstructured data via **Cognitive Search**, and deployed **full-stack** ML solutions with Node.js and Azure services
- Architected MLOps framework with Azure **DevOps**, MLflow, and Infrastructure as Code for one-click model deployments, automated drift detection, and event-driven microservices on Azure **Kubernetes** Service with Cosmos DB monitoring

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R, Shell - Scripting

Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI

Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, Microsoft Office, Power BI

Libraries: pandas, NumPy, Matplotlib

Awards: Won **"Rookie Of The Year"** Award for Client Services, Won Top 8 ideas for UNHACK 2023

Relevant Courses: Calculus, Net-Centric Web Development/Mobile Computing, Object-oriented programming, Data Structures & Algorithms, Computer Network and Protocols, Malware Analysis, Applied Cryptography, Full-Stack, Operating Systems, TCP/IP - Protocols/layers